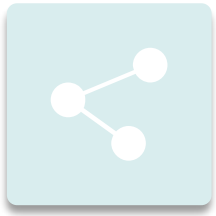
NORTHEASTERN UNIVERSITY

Artificial Intelligence Technology Project

Instructor： 于瑞云
College： Software College
Submission Email： 2940986065@qq.com

Deadline： November 15th



Pascal VOC2012 Semantic Segmentation

In this project, you will make semantic segmentation predictions for images. In order to do that, you need to build a semantic segmentation model, train on the Pascal VOC2012 data set, and inference on the given test set. In this process, you need to learn to load data, build models, train and validate models, build loss functions, etc.

You will also need to write a report on what you learned during the project. Submit your works as required before the deadline. Any plagiarism or forgery will be considered as a “fail” for the project. We have established a complete inspection process, please do not touch the red line.





Pascal VOC2012 Dataset

Dataset1 download link:
http://host.robots.ox.ac.uk/pascal/VOC/voc2012/index.html#devkit

Development Kit

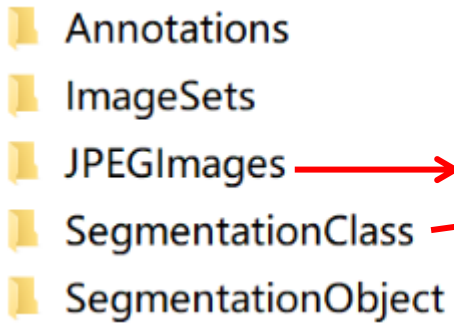
The development kit consists of the training/validation data, MATLAB code for reading the annotation data, support files, and example implementations for each competition.

The development kit is now available:

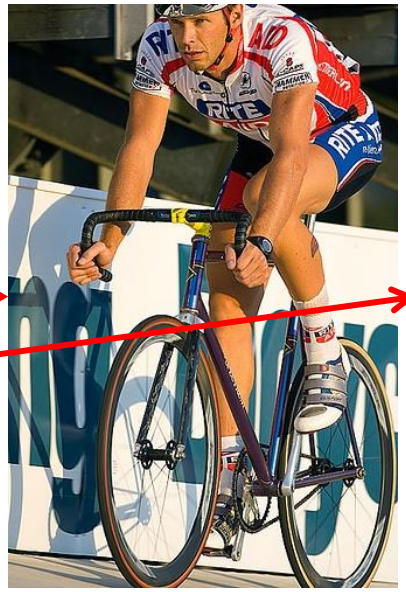
- Download the training/validation data (2GB tar file)
- Download the development kit code and documentation (500KB tar file)
- Download the PDF documentation (500KB PDF)
- Browse the HTML documentation
- View the guidelines used for annotating the database (VOC2011)
- View the action guidelines used for annotating the action task images

Dataset Content:

Segmentation set consists:
train set: 1465 images,
validation set: 1450 images
(optional)



Image



Prediction



Dataset2 download link:
https://pan.baidu.com/s/1Y5NhhXa67kstk7nHdGbFgw?pwd=tnam
Extraction code: tnam

Dataset2 is utilized as the test set to evaluate the model.
test set: 400 images (extra set, not belonging to Pascal; only images, no label)

Task

1. Load segmentation data from the dataset, including images, labels, etc.
2. Construct the semantic segmentation model. Ensure that the input and output sizes of adjacent layers are compatible.
3. Choose the appropriate loss function and build the backpropagation.
4. Train and validate(optional) the model with Pascal VOC2012 segmentation train set and validate set(optional).
5. Save the weights of the model.
6. Perform inference on the test set, and save the outputs which are single channel gray scale masks.
7. The entire training process needs to be visualized.(use tensorboard)
8. Write the report.

Submission

- A project report (e.g. report_studentname_studentId.docx)
  1. The data processing and loading methods, the principle of model design, and the method of adjusting the parameters
  2. Figure of network model structure
  3. Figures for the accuracy curves and loss curves

- Your source code file. (e.g. code_studentname_studentid)

- The prediction of test set. (e.g. predict_studentname_studentid.zip)

- Zip these three files into a folder named studentname_studentid, and submit the zipped folder.

Note:

- ✓ Final score is based on mIoU performance of the model on the test dataset and quality of the experimental report.
- ✓ No plagiarism. It is very easy to check whether the predictions are plagiarized.
- ✓ Designing a new network model and using validation set to validate the model after each training epoch are encouraged.

Your project should be submitted to the following email address 2940986065@qq.com by November 15th.

Late submission leads to “fail” for the project.